

Mechanics WS 1

In the following Q1-8, take $g = 10 \text{ m.s}^{-2}$

1. The combined air and road resistance of a car in motion is proportional to v^2 , where v is its speed. When the engine is disengaged the car moves down an incline making an angle $\sin^{-1}\left(\frac{1}{30}\right)$ with the horizontal, with a velocity of 30 m/s. find the force required to drive the car up the incline with a steady speed of 24 m/s, given that the mass of the car is 1200 kg.
2. A truck of mass M is driven up a road inclined at an angle θ to the horizontal. After its speed has reached u m/s the engine continues to exert a constant force of F newtons. If the resistance R is constant, find the time taken to reach the velocity v m/s.
3. A car of mass 1500 kg is moving at 60 km/h; when the brakes are applied with a braking force of 12000 N.
Find:
 - a. The acceleration
 - b. The time taken by the car to stop
 - c. The distance travelled before coming to rest.
4. A body of mass m is pulled up a smooth incline making an angle θ with the horizontal and has an acceleration f . Find the force F that pulls the body.
5. A mass of 10 kg is pulled along the horizontal by a chain making an angle of 30° with the horizontal. If the tension in the chain is 50 N, find the acceleration of the body and the magnitude of the normal reaction.
6. A smooth block of mass 2 kg slides down an incline making an angle of $\tan^{-1}\left(\frac{3}{4}\right)$ with the horizontal. If the tension in the chain is 50 N, find the acceleration of the body and the magnitude of the normal reaction.
7. A truck of mass 2000 kg starts to climb an incline of angle given by $\theta = \sin^{-1}\left(\frac{1}{10}\right)$. The total resistive force is 2000 N. Find the retardation it experiences.
8. Find the magnitude of the braking force to stop a car of mass 1200 kg in 20 m when it is travelling at 60 km/h
 - a. On a horizontal road
 - b. Down an incline of an angle $\sin^{-1}\left(\frac{1}{40}\right)$.
9. The rise and fall of the tide at a certain port may be considered as simple harmonic, the time difference between successive high tides being 10 hours. The harbour entrance has a depth of 20 m at high tide and 8 m at low tide. If the low tide occurs at 10.00 a.m. on a certain day, find the earliest time that a cargo ship requiring a minimum depth of 15 m of water can pass through the entrance.
10. The acceleration of body moving along the x - axis is given by
$$\frac{d}{dx}\left(\frac{v^2}{2}\right) = -x \text{ for } x \leq 1$$
$$= 2 - x \text{ for } x > 1.$$
 - a. Show that the quantity $\frac{1}{2}\left(\frac{dx}{dt}\right)^2 + f(x)$ is a constant of motion, where $f(x) = \frac{x^2}{2}$ or $\frac{1}{2}(x - 2)^2$ according as $x \leq 1$ or $x > 1$.
 - b. When $t = 0, x = 0$ and the initial velocity of the body is $1/3$ in the positive direction. Determine the extreme points of the subsequent motion, hence prove that the motion is S.H.M.
 - c. If in Exercise (b), $v = 2$, find the extreme points of the motion. Explain whether the motion now is SHM or not.
11. Show that $x = r \cos(\omega t + \phi)$ where r, ω, ϕ are constants, is a solution of $\frac{d^2x}{dt^2} = -\omega^2 x$. A small naval target rises and falls with SHM of period 10 seconds; the height of the waves from the crest to trough is 2 m. At a horizontal range of 2000 m a gun is fired so that the target would be hit provided it remains stationary in its highest position. The horizontal component of velocity is 1000 m/s.

Show that the target would be missed by vertical height of approximately 0.69 m.

12. A particle moves in a straight line Ox and its equation of motion is $\frac{d^2x}{dt^2} = 9 - 9x$. Prove that the displacement at time t is given by $x = 1 + \frac{5}{3} \cos(3t - 127^\circ)$, given that $v = 4 \text{ m/s}$, when $x = t = 0$. What is the centre of oscillation? Find the amplitude and the maximum velocity.

13. A particle is projected from the origin with initial velocity U to pass through a point (a, b) . Prove that there are two possible trajectories if $(U^2 - gb)^2 > g^2(a^2 + b^2)$

14. A particle of mass 10 kg is suspended by two strings of length 3 m and 4 m attached to two points at the same level 5 m apart. Find the tensions in the strings. Take $g = 10 \text{ m.s}^{-2}$

15. A particle moves according to the equation $v^2 = 16 - 4x^2$ where x is its position in metres and v is its velocity in metres per second.

Initially the particle is at the origin and moves off in a positive direction

- Prove the particle has acceleration in the form $a = -n^2x$ where n is a constant
- Find the amplitude of the motion, explaining how this is found
- Use integration to find an expression for x as a function of time t beginning with $v = \sqrt{16 - 4x^2}$

16. The displacement x metres of a particle after t seconds, is given by $x = 5 \sin 3t - 7 \cos 3t$

- Show that the motion of the particle is SHM
- Find the maximum displacement
- Find the time when the particle first passes through the centre of motion
- Sketch the graph of the acceleration versus displacement

17. The acceleration of a particle moving along a straight line is given by $a = 3x(x - 2)$, where x metres is its displacement from the origin O after t seconds. Initially, it is at O and its velocity is 2 m/s.

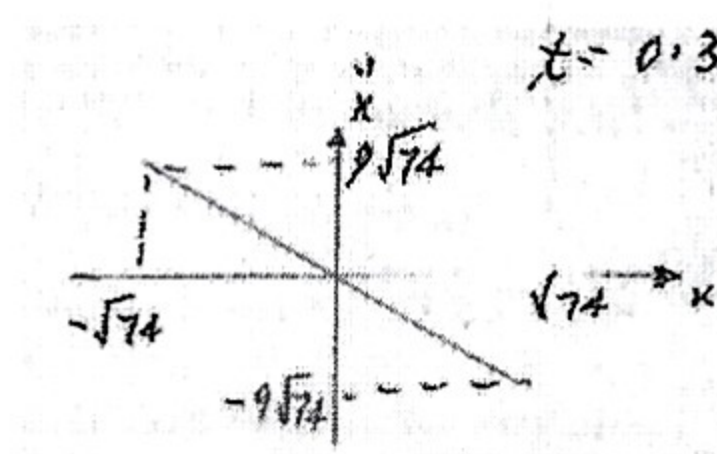
- Show that $v = 2(x^3 - 3x^2 + 2)$, where v is its velocity.
- Find its velocity and acceleration at $x = 1$.
- Briefly describe its motion after it moves from $x = 1$.

18. The acceleration of a particle P is given by the equation $\ddot{x} = 8x(x^2 + 1) \text{ ms}^{-2}$, where x is the displacement of P from the origin in metres after t seconds, with movement being in a straight line. Initially the particle is projected from the origin with a velocity of 2 ms^{-1} .

- Show that the velocity of the particle can be expressed as $v = 2(x^2 + 1)$.
- Hence, show that the equation describing the displacement of the particle at time t is given by $x = \tan 2t$.
- Determine the velocity of the particle at time $\frac{\pi}{8}$ seconds.

Answer:

- 656N
- $\frac{M(v-u)}{F-R-10M \sin \theta}$
- (a) -8 m.s^{-2} (b) 2.08s (c) 17.4m
- $F = m(f + \sin \theta)$
- $a = 4.33 \text{ m.s}^{-2}$, Reaction 75N.
- $a = -6 \text{ m.s}^{-2}$, Reaction 16N.
- -2 m.s^{-2}
- a. 8330N; b. 8030N
- 12.46 p.m
- (b) $x = \pm \frac{1}{3}$ (c) $2 \pm 2\sqrt{2}$, motion is not S.H.M.
- Centre at $x = 1$, Amplitude = $\frac{5}{3}$, Max. velocity 5 m/s
- $T_1 = 80 \text{ N}, T_2 = 60 \text{ N}$
- a. $\ddot{x} = -2^2x$; b. $A = 2 \text{ metres}$; c. $x = 2 \sin 2t$
- $\sqrt{74}, t = 0.32$,



17. b. $v = 0, \ddot{x} = -3 \text{ m/s}^2$
18. c. 4 ms^{-1}